

Midterm 2 Review

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Exercise 1. We call an operator **normal** if it commutes with its hermitian conjugate, $[A, A^\dagger] = 0$. In particular, hermitian, anti-hermitian, and unitary operators are all normal, so this is a more general class of operators than hermitian operators.

- Show that if A is normal and $A|u\rangle = a|u\rangle$ for some nonzero $|u\rangle$, then $A^\dagger|u\rangle = a^*|u\rangle$.
- Show that (a) implies that the eigenbras of A are the hermitian conjugates of the eigenkets, and the left spectrum is identical to the right spectrum.
- Show that the eigenspaces corresponding to distinct eigenvalues of a normal operator are orthogonal. This is a generalization of the proof for hermitian operators.

Exercise 2. Unitary Operators. An operator U is called **unitary** if $UU^\dagger = 1$. This problem shows that in an infinite dimensional Hilbert space, if an operator preserves the norm of all vectors and possesses both a left and right inverse, then it is unitary.

- Let U be some operator on a finite-dimensional vector space, and let $|\psi'\rangle = U|\psi\rangle$. Show that $\langle\psi'|\psi'\rangle = \langle\psi|\psi\rangle$ if and only if U is unitary.
- Consider the Hilbert space for a particle moving in one dimension (which is infinite-dimensional). Let $|n\rangle$ be the usual harmonic oscillator eigenbasis, with $n = 0, 1, \dots$. Define an operator U by

$$U|n\rangle = |n+1\rangle.$$

Show that if $|\psi'\rangle = U|\psi\rangle$, then $\langle\psi'|\psi'\rangle = \langle\psi|\psi\rangle$. Then show that

$$U^\dagger U = \mathbb{1},$$

so that U has a left inverse, and it is $U^\dagger$. Does U have a right inverse? Is U unitary?

Exercise 3. Some More Unitary Operators. Let H be a hermitian operator and $U = e^{iH}$.

- Show that U is unitary. (Notice the analogy with scalars: If θ is real, then $e^{i\theta}$ has unit modulus.)
- If U_1, U_2 are unitary operators, show that $U_1 U_2$ is unitary. If $U_1 U_2$ is unitary, does that imply that U_1 and U_2 are?
- Challenge.** Show that $\det(U) = e^{i \operatorname{tr}(H)}$.

Exercise 4. More Fun with Pauli Matrices. Consider the 2×2 matrices,

$$\mathbb{1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The matrices $\sigma_x, \sigma_y, \sigma_z$ are called the **Pauli matrices**; we will also denote them $\sigma_1, \sigma_2, \sigma_3$.

a) Show that the result from homework 5,

$$\sigma_i \sigma_j = \delta_{ij} + i \varepsilon_{ijk} \sigma_k, \quad (1)$$

implies that

$$[\sigma_i, \sigma_j] = 2i \varepsilon_{ijk} \sigma_k$$

and

$$\{\sigma_i, \sigma_j\} = \sigma_i \sigma_j + \sigma_j \sigma_i = 2\delta_{ij}.$$

b) What we mean by a **vector operator** is a vector of operators; for example, $\mathbf{A} = (A_x, A_y, A_z)$, where A_x, A_y , and A_z are operators. Two vector operators are said to commute if all of their components commute. Show that, in general, $\mathbf{A} \cdot \mathbf{B} \neq \mathbf{B} \cdot \mathbf{A}$ and $\mathbf{A} \times \mathbf{B} \neq -\mathbf{B} \times \mathbf{A}$.

c) Let $\hat{n}$ be an arbitrary unit vector and θ an arbitrary angle. Show that

$$\exp(-i\theta \boldsymbol{\sigma} \cdot \hat{n}) = \mathbb{1} \cos \theta - i(\boldsymbol{\sigma} \cdot \hat{n}) \sin \theta.$$

We often drop the $\mathbb{1}$, it being understood that a scalar like $\cos \theta$ is multiplied by the identity matrix.

Hint: The power series for sine and cosine will be useful here.

d) Show that if A and B are square matrices, then

$$\text{tr}(AB) = \text{tr}(BA).$$

e) Show that $\text{tr}(\sigma_i) = 0$. Then, use Equation 1 to find a simple expression for $\text{tr}(\sigma_i \sigma_j)$.

f) The set of four 2×2 matrices $\{\mathbb{1}, \boldsymbol{\sigma}\}$ forms a basis in the space of 2×2 matrices; that is, an arbitrary 2×2 matrix can be expressed as a linear combination of these four matrices. Thus, if M is some 2×2 matrix, we can always write

$$M = a\mathbb{1} + \mathbf{b} \cdot \boldsymbol{\sigma},$$

which follows from the fact that $\mathbb{1}$ and the Pauli matrices are linearly independent, as can be seen by just staring at the four matrices. Find nice expressions for the expansion coefficients a and $\mathbf{b} = (b_1, b_2, b_3)$. Do this by taking traces or by multiplying by a Pauli matrix and then taking traces. Show that M is hermitian if and only if a and b_i are real.

g) Express the following as linear combinations of the Pauli matrices and $\mathbb{1}$.

i) $(\mathbb{1} + i\sigma_x)^{1/2}$

Hint: Relate it to a certain half-rotation.

ii) $(2\mathbb{1} + \sigma_x)^{-1/2}$.

iii) σ_x^{-1} .

h) Suppose that M is only nonzero in the (r, s) index, where it has a 1. That is, suppose we can write

$$M_{ij} = \delta_{ir}\delta_{js}.$$

Use this to find a nice expression for $(\sigma_m)_{ij}(\sigma_m)_{kl}$, where we again use the convention that the m index is summed over, in terms of Kronecker deltas.

Exercise 5. Even More Fun with Pauli Matrices. In this exercise, you will give an explanation of the following fact from Week 5 Worksheet 2. Let $\hat{n}$ be a unit vector. By Exercise 2(a) of the same worksheet, $\hat{z}$ can be rotated into $\hat{n}$ using just two angles. Let R be the (3×3) -matrix which rotates $\hat{z}$ into $\hat{n}$, i.e. $R\hat{z} = \hat{n}$. Now, let $U = e^{-\frac{i}{2}\alpha\sigma_z}e^{-\frac{i}{2}\beta\sigma_y}$, then

$$U^\dagger \boldsymbol{\sigma} U = R\boldsymbol{\sigma},$$

where on the RHS we understand R as rotating the components of the *vector* $\boldsymbol{\sigma}$, just as it would rotate the components of a vector $\mathbf{a}$ whose entries were scalars.

a) Use Exercise 4(c) to calculate $U^\dagger \boldsymbol{\sigma} U$.

b) Prove the identities

$$[\hat{n} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] = -2i\hat{n} \times \boldsymbol{\sigma}, \quad (\hat{n} \cdot \boldsymbol{\sigma})(\hat{n} \cdot \boldsymbol{\sigma}) = 2\hat{n}(\hat{n} \cdot \boldsymbol{\sigma}) - \boldsymbol{\sigma}.$$

c) Use (b) to rewrite your result from (a) as

$$U^\dagger \boldsymbol{\sigma} U = \cos(\theta)\boldsymbol{\sigma} + (1 - \cos \theta)\hat{n}(\hat{n} \cdot \boldsymbol{\sigma}) + \sin(\theta)\hat{n} \times \boldsymbol{\sigma}.$$

d) **Challenge.** Give a physical explanation for why the result of (c) is reasonable for rotations, and why it should be the same as $R\boldsymbol{\sigma}$.

Exercise 6. Ignore the fact that the hydrogen atom is a three-dimensional problem, and imagine that

$$H = \frac{p^2}{2m} - \frac{e^2}{r}$$

corresponds to a one-dimensional problem, where $p^2 = p_x^2 + p_y^2 + p_z^2$ and $r^2 = x^2 + y^2 + z^2$. Assuming that

$$\Delta p \Delta r \geq \hbar/2,$$

estimate the ground state energy. (The actual ground state energy is about -13.6 eV).

Hint: To make your calculation easier, recall that $mc^2 = 511$ keV is the electron rest energy. You can use that the Bohr radius is

$$a_0 = \frac{\hbar^2}{me^2} \approx 137 \cdot \frac{\hbar}{mc}.$$